

Pattern Matching and Machine Learning for Audio Signal Processing

Exercise sheet 2

To be uploaded in eCampus till: 02-05-2026 22:00 (strict deadline)

Exercise 2.1

[4 points]

Let $1 \neq q \in \mathbb{C}$. Prove that for all positive integers N

$$\sum_{n=0}^{N-1} q^n = \frac{1 - q^N}{1 - q}.$$

Exercise 2.2

[9 + 3 = 12 points]

For a fixed positive integer N , we consider two particular bases in the space $\mathbb{C}^N$ of length N signals:

- the ON-basis $\mathbf{e}_0, \dots, \mathbf{e}_{N-1}$ of unit vectors (representing an extremely local view) and
- the unnormalized Fourier basis $\mathbf{u}_0, \dots, \mathbf{u}_{N-1}$ (representing an extremely global view).

Let $\rho := \exp(2\pi i/N)$ and $\omega := \exp(-2\pi i/N)$ denote two primitive N th roots of unity. Recall that $\mathbf{u}_k(n) := \exp(2\pi i k n/N) = \rho^{kn}$, for $k, n \in [0 : N-1]$. Furthermore, the $N \times N$ DFT matrix DFT_N is defined by $\text{DFT}_N(k, n) := \omega^{kn}$, for $k, n \in [0 : N-1]$.

(a) Prove that for $k \in [0 : N-1]$ the following holds:

- (i) $\overline{\mathbf{u}_k} = \mathbf{u}_{N-k}$.
- (ii) $\text{DFT}_N \mathbf{e}_k = \mathbf{u}_{N-k} = \overline{\mathbf{u}_k}$.
- (iii) $\text{DFT}_N \mathbf{u}_k = N \cdot \mathbf{e}_k$.

(b) Explicitly write down DFT_3 . Simplify the matrix entries as much as possible.